Seth Schwartz

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PROFESSIONAL SUMMARY

Senior Pipeline Engineer and technical leader with 14+ years designing and scaling 3D content, real-time, and animation pipelines across feature film, interactive, and automotive visualization. Builds high-impact automation that cuts production timelines, improves data consistency, and frees artists for creative work. Experienced leading cross-functional teams and aligning technical systems with creative and business goals. Credited on 10 feature films. Currently building Roxy, a procedural DCC with OpenUSD, a GPU path tracer, and an LLM agent that drives the tool through its own command layer, with a headless harness for evaluating agents on modeling tasks.

CORE COMPETENCIES

- **Pipeline Development:** Animation & Asset Pipelines, Automation, Tool Design, Workflow Optimization
- **Production Infrastructure:** Source Control (Git, Perforce), Asset Versioning, Vendor Data Integration
- **Leadership:** Technical Leadership, Mentorship, Cross-Departmental Collaboration
- **Languages:** Python, C++, MEL, GDScript
- **Tools & Engines:** Autodesk Maya, Houdini, Unreal Engine, Godot, OpenUSD, Qt/PySide6, Taichi, Deadline Cloud
- **Systems:** GPU Path Tracing, LLM Agents (Claude API), Agent Evaluation, Pipeline Architecture, Troubleshooting

PROFESSIONAL EXPERIENCE

Senior Pipeline Engineer
Team One — September 2025 – Present

- Architected a fully automated pipeline for ingesting Lexus vehicle data from Toyota Motor Company into Autodesk Maya, building Unreal-ready assets for vehicles of thousands of parts each.
- Developed high-performance automation tooling that reduced a 4-week manual build process to under 5 minutes, enabling rapid iteration at scale and significantly reducing artist downtime.
- Designed and implemented a vehicle configuration pipeline supporting part and material variants configurable to dealer requirements while enforcing TMC-approved build constraints.
- Automated the pipeline end to end, from data ingest through Unreal Engine import to final renders, and established the Deadline Cloud render farm it runs on, shifting artists toward bespoke client work.
- Driving adoption of OpenUSD for vehicle master builds, designing composition strategies (references, payloads, variants, instancing) to increase performance and substantially reduce data footprint.

Senior Pipeline Technical Director (Contract)
Duolingo — April 2025 – June 2025

- Built a 3D animation pipeline from the ground up for a next-generation app feature, integrating Maya with the Godot game engine.
- Designed and implemented custom publishing tools and APIs consumed by artists and engineers, standardizing asset and animation export between DCC and engine.
- Established data structures, versioning, and automation systems ensuring scalability and consistency across production assets.
- Built batch processing, conversion, and validation tooling that significantly reduced manual workload and error rates.
- Mentored junior technical artists and animators on best practices, tool usage, and technical problem-solving.

Lead Pipeline Technical Director
Reel FX Animation Studios — November 2019 – April 2025

- Led development of scalable Python pipeline tooling supporting VFX, animation, and post-production workflows for feature films and episodic content.
- Established shared frameworks, coding standards, and reusable patterns adopted across production departments, raising the technical bar for tool development studio-wide.
- Designed and implemented remote collaboration tooling during the pandemic, enabling uninterrupted distributed production across concurrent projects.
- Conducted strategic pipeline analysis and planning, evaluating new technologies and defining long-term infrastructure investments.
- Partnered with department leads to identify workflow bottlenecks, driving measurable efficiency gains across production stages.
- Oversaw migration and integration of external vendor data, ensuring reliable transitions into proprietary formats.

Animation Technical Director
Reel FX Animation Studios — October 2012 – November 2019

- Developed and maintained custom tools for the animation, layout, and environment teams, ensuring efficiency and precision in production workflows.
- Established previz and layout pipelines and built inter-departmental tools to enhance cross-functional collaboration and productivity.

SELECTED PROJECTS

Roxy — Procedural DCC (personal R&D)
roxy.sethschwartz.dev · Python / PySide6 / Qt RHI / Taichi

- Built an embedded assistant on the Claude API that models, shades, and renders through the same commands as the UI. It has eight tools, receives the scene diff each turn, and gates spending actions behind an approval click with per-turn and monthly caps. It builds a Cornell box from prompt to saved render in about 5 minutes; each turn is a single undo step.
- Built a headless evaluation harness for AI agents: modeling tasks run in parallel without the UI, every command is traced to a replayable log, and results are scored on measured geometry (topology, validity, scale) and render fit against reference drawings.
- Building a dataset of modeling operations paired with their measured effects (topology, geometry, shading, UV), combined with technique records from artist tutorials, to improve agent modeling.
- Wrote a GPU path tracer from scratch (Taichi on Metal and CUDA): Principled BSDF, MIS, two-level BVH, subsurface scattering, photon-mapped caustics, and a guided denoiser. Sustains 140M rays/s at 1080p on an M1 Max and 681M on an RTX 5090.
- Architected a Python command layer of 279 commands in 23 categories that every edit routes through: Qt UI, scripts, and the assistant. Undo is grouped commands, so a gizmo drag, script call, or assistant turn is one step.
- Built an OpenUSD importer that resolves up-axis, unit, and matrix conventions and reports every lossy conversion; imports Pixar's Kitchen_set (1,788 meshes, 536,949 triangles) in about 1.4 s. Loads Gaussian splat captures (305,872 splats in 0.08 s) as selectable scene objects.
- Procedural modelling through per-shape node graphs (100+ node types, including a VEX-style wrangle compiled to GPU kernels). 15,000+ automated tests; 565 merged PRs over 5.5 months.

EDUCATION

M.S., Visualization Science
Texas A&M University, May 2015 · Thesis: Procedural Animation for Tentacled Digital Creatures

B.S., Environmental Design, Minor in Mathematics
Texas A&M University, May 2008